

Faris Maulana

AI Engineering Manager | Applied AI Engineer | LLM Systems | Data and Software Architecture

I build AI systems around real operational problems. That usually means starting with the data, understanding how people already work, and then deciding where an AI model, an agent, or a conventional application can genuinely remove friction.

Enterprise AI

Agents and decision support

LLM systems

RAG, evaluation and Text-to-SQL

Data platforms

SQL, DWH and analytics

Software

APIs and workflow applications

Security

RBAC, auditability and red-teaming

Location

Bogor / Jakarta

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Portfolio: <https://faris-portfolio-red.vercel.app/>

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About Me

I am an AI and data engineer based in Bogor and Jakarta. My work has moved from data analysis and business intelligence into enterprise AI architecture, LLM applications, and the data platforms that support them. I am most useful in projects where the technical problem is tied closely to an operational workflow: network monitoring, management reporting, appraisal processes, customer retention, or access to information spread across multiple systems.

I currently lead AI engineering work at PT Trans Indonesia Superkoridor. The role involves more than selecting a model. I am helping establish the data foundation, the architecture for on-premise model serving, and the agent layer that will eventually let users ask operational questions in natural language. In parallel, my consulting work has involved prompt design, LLM evaluation, RAG, fine-tuning concepts, RLHF workflows, and production-readiness discussions.

How I Work

- Start with the business process and the source systems, not with a chatbot interface.
- Make data lineage, access control, testing, and observability part of the design from the beginning.
- Use AI only where probabilistic behavior is acceptable and add validation where it is not.
- Keep architecture practical enough for the available infrastructure, team, budget, and operating model.
- Measure the outcome in operational terms: time saved, reporting effort reduced, better access to information, or clearer decisions.

Current Focus

- Enterprise Text-to-SQL and multi-agent orchestration
- On-premise LLM serving and model evaluation
- RAG quality, retrieval design, reranking, and grounded answers
- Data warehouse architecture for AI consumption
- AI security, red-teaming, guardrails, access control, and auditability
- Reliable AI-assisted software development workflows

Professional Experience

Manager, AI Engineering | PT Trans Indonesia Superkoridor

Period: May 2026 - Present

I am building the AI engineering function and the technical foundation required for enterprise AI use cases in a national fiber-network environment. A major part of the work is understanding where operational data lives, how it should move from source systems into governed analytical layers, and what needs to be in place before an agent can safely query it.

- Designed a medallion-style data platform using ClickHouse, PostgreSQL, and dbt-core.
- Defined ingestion and migration paths across MySQL and Ubuntu-based systems.
- Designed a LangGraph multi-agent approach for Text-to-SQL, including planning, generation, validation, and response stages.
- Evaluated on-premise serving options including vLLM and SGLang.
- Worked with operational and security stakeholders on deployment boundaries, access, and validation.

Master Consultant, AI Engineer | iMerit Technology

Period: December 2025 - Present, according to the supplied CV

My work is centered on the practical reliability of LLM systems: selecting APIs, improving prompts, evaluating outputs, thinking through fine-tuning and RLHF workflows, and helping shape RAG and governance patterns for production use.

Data Analyst, AI/BI | PT Berlian Laju Tanker Tbk

Period: December 2024 - May 2026

I worked across Finance, Operations, Technical, Safety, and HR. The role combined SQL, Oracle, Power BI, forecasting automation, and application development. I built more than 20 dashboards, improved access to Danaos ERP data, and developed a multi-role crew appraisal application.

Earlier Applied AI and Data Science Work

Earlier roles and engagements at Toyota Research Institute and Telkom Indonesia involved predictive modeling, NLP, computer vision, REST APIs, stakeholder communication, and the translation of research or model outputs into business recommendations.

Selected Projects

1. Enterprise Text-to-SQL Platform

Problem and goal: The goal is to let operational and management users ask questions about fiber-network operations without requiring them to know the database structure. The difficult part is not generating SQL. It is making sure that the agent uses the right source, respects access boundaries, understands the schema, and can explain what it did.

What I designed and built: I designed the data and agent architecture around ClickHouse, PostgreSQL, dbt-core, Airflow, Airbyte, OpenMetadata, LangGraph, vLLM, and SGLang. The plan separates schema discovery, question planning, SQL generation, validation, execution, and answer synthesis.

Outcome: The platform is still being developed. The expected outcome is a controlled path from OSS data to traceable natural-language decision support, with production data sourced from OSS through the DWH rather than from spreadsheet files.

2. NOC Monitoring AI Agent

Problem and goal: NOC teams need fast access to incident and telemetry information, especially when the user asking the question does not know the table structure or query language.

What I designed and built: I developed a proof of concept using FastAPI, LangChain, ClickHouse, and WhatsApp integration through Fonnte. The design combined natural-language incident questions with alert delivery.

Outcome: The proof of concept showed how conversational access could sit on top of network data and support faster triage. Further production work would require stronger authorization, alert rules, observability, and integration testing.

3. RAG System and Evaluation Framework

Problem and goal: I wanted to build a complete RAG pipeline rather than a demo that only retrieves documents and sends them to a model.

What I designed and built: The project ingests more than 50 AI and machine-learning articles, creates embeddings, stores them in ChromaDB, reranks results with a cross-encoder, generates answers through Groq, audits retrieval diversity, and includes embedding fine-tuning and evaluation components.

Outcome: The result is a modular system covering ingestion, retrieval, reranking, generation, evaluation, and an interactive Gradio interface. It also gave me a practical basis for discussing faithfulness, relevance, bias, and retrieval quality.

4. Maritime Performance Appraisal Application

Problem and goal: The existing appraisal process involved several roles and approval stages. A simple sequential workflow would create delays whenever one appraiser had not completed a task.

What I designed and built: I designed RBAC for appraisees, supervisors, reviewers, and decision makers, then implemented a parallel state-machine approach so that peers in the same stage could work independently. The system also included weighted scoring, audit logs, and Playwright end-to-end testing.

Outcome: The application turned a fragmented process into a traceable workflow and provided a clearer view of progress, permissions, and scoring across the appraisal lifecycle.

Analytics and Decision-Support Work

Maritime fleet analysis

I analyzed 36 months of fleet data to understand how maintenance compliance, voyage decisions, safety, and commercial performance moved together. The portfolio analysis used PCA and multivariate regression. It reported that two components explained 82% of the variance in fleet profitability, and estimated associations of 8.5% lower OPEX volatility with improved maintenance compliance and 11.2% higher TCE yield with ballast-speed optimization. These figures should be read as results within the analyzed dataset, not universal causal claims.

B2B churn and customer lifetime value

Using an 18-month cohort of 5,000 enterprise accounts, the portfolio applied Kaplan-Meier survival analysis and regularized logistic regression. It identified early AI-module adoption and API usage as strong retention signals, with a modeled threshold around 10,000 API calls per month.

Clinical triage workflow

The analysis examined when ML-supported triage saves time and when false positives create more review work than the model removes. The reported model suggested an 88% confidence gate and showed that false-positive rates above 5% could erase the workflow benefit.

Learning and retention analytics

A mixed-effects model and a decision tree were used to compare active coding with passive video learning and to examine connectivity constraints. The portfolio reported a stronger graduation effect for sandbox activity, an 850 ms latency breakpoint, and a 14-day early-warning window.

Enterprise BI

At PT Berlian Laju Tanker, I built more than 20 Power BI dashboards and used Oracle SQL, Power Query, and VBA to reduce recurring manual reporting. The supplied CV reports a 50% reduction in manual decision-support reporting.

Earned Certifications and Completed Learning

BNSP Certificate of Competence: Lead Programmer

Issuer: BNSP through LSP Universitas Pancasila

Completed / issued: 30 November 2022

Validity: Valid for three years; expired 30 November 2027

Type: Professional competency certification

This is the strongest formal software-development credential in the portfolio. The assessment covered 21 competency units, including application architecture, scalability, network and real-time programming, parallel programming, code review, UAT, cutover, monitoring, migration, static testing, and information-security risk and standards.

Source: Credential evidence - Uploaded BNSP certificate No. 62019 2514 6 0001233 2022

Bachelor of Engineering in Informatics

Issuer: Universitas Pancasila

Completed / issued: Judicium 22 February 2024; diploma dated 16 May 2024

Validity: No expiration

Type: Academic degree

The diploma confirms completion of the undergraduate Informatics Engineering program and the award of Sarjana Teknik (S.T.). This wording follows the diploma rather than the older CV date.

Source: Credential evidence - Uploaded diploma No. 552012024000131

Introduction to Data Analytics

Issuer: Provider not visible on the supplied certificate

Completed / issued: 3 December 2025

Validity: No expiration stated

Type: Course completion certificate

A completed introductory program in data analytics. The certificate displays code 9517352.

Source: Credential evidence - Uploaded certificate code 9517352

Data Science Fundamentals

Issuer: Dicoding Indonesia

Completed / issued: 23 November 2023

Validity: Valid until 23 November 2027

Type: 11-hour assessed course

Credential 0LZ0QKM13Z65; data fundamentals, analysis lifecycle, SQL/NoSQL, tools, introductory ML and portfolio development.

Source: Credential evidence - <https://www.dicoding.com/certificates/0LZ0QKM13Z65>

SQL Fundamentals

Issuer: Dicoding Indonesia

Completed / issued: 30 November 2023

Validity: Valid until 30 November 2027

Type: 11-hour assessed course

Credential KEXL8EWR0ZG2; relational databases, DBMS, DDL, DML, constraints, keys, indexes and basic queries.

Source: Credential evidence - <https://www.dicoding.com/certificates/KEXL8EWR0ZG2>

Google Cloud Fundamentals

Issuer: Dicoding Indonesia

Completed / issued: 31 May 2022

Validity: Expired 31 May 2027

Type: 12-hour assessed course

Credential NVP7KKDQVZR0; Compute Engine, VPC, regions, storage, databases, IAM, encryption, compliance and cost models.

Source: Credential evidence - <https://www.dicoding.com/certificates/NVP7KKDQVZR0>

Data Visualization Fundamentals

Issuer: Dicoding Indonesia

Completed / issued: 24 June 2022

Validity: Expired 24 June 2027

Type: 16-hour assessed course

Credential 6RPNDV9Y8Z2M; Google Sheets, data preparation, storytelling, Gestalt principles and preattentive attributes.

Source: Credential evidence - <https://www.dicoding.com/certificates/6RPNDV9Y8Z2M>

Claude Code in Action

Issuer: Anthropic

Completed / issued: 23 March 2026

Validity: No expiration stated

Type: Certificate of completion

Verified certificate aqfbcjyv6a88; AI-assisted software development and Claude Code learning.

Source: Credential evidence - <https://verify.skilljar.com/c/aqfbcjyv6a88>

Claude 101

Issuer: Anthropic

Completed / issued: Date not displayed

Validity: No expiration stated

Type: Certificate of completion

Foundational Claude learning.

Source: Credential evidence - Uploaded certificate

Introduction to Claude Cowork

Issuer: Anthropic

Completed / issued: Date not displayed

Validity: No expiration stated

Type: Certificate of completion

Collaborative AI workflow learning.

Source: Credential evidence - Uploaded certificate

Data for Breakfast Virtual, Asia Pacific

Issuer: Snowflake

Completed / issued: 13-17 April 2026

Validity: No expiration stated

Type: Certificate of attendance

Participation in Snowflake professional development on modern data platforms.

Source: Credential evidence - Uploaded certificate

Python

Issuer: Kaggle Learn

Completed / issued: 6 April 2022

Validity: No expiration stated

Type: Course completion

Programming foundation later applied to data, ML, APIs and LLM systems.

Source: Credential evidence - Uploaded certificate

Data Cleaning

Issuer: Kaggle Learn

Completed / issued: 6 April 2022

Validity: No expiration stated

Type: Course completion

Practical data-quality foundation.

Source: Credential evidence - Uploaded certificate

Data Visualization

Issuer: Kaggle Learn

Completed / issued: 8 April 2022

Validity: No expiration stated

Type: Course completion

Visualization foundation later applied to business dashboards.

Source: Credential evidence - Uploaded certificate

Azure AI Apps and Agents Developer Associate (AI-103)

Provider: Microsoft

Issuer: Microsoft Learn

Completed / issued: 6 April 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://learn.microsoft.com/en-us/credentials/certifications/azure-ai-apps-and-agents-developer-associate/>

AWS Certified Machine Learning Engineer - Associate

Provider: AWS

Issuer: AWS Certifications

Completed / issued: 7 Jan 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://aws.amazon.com/certification/>

AWS Certified Generative AI Developer - Professional

Provider: AWS

Issuer: AWS Certifications

Completed / issued: 10 March 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://aws.amazon.com/certification/>

Professional Machine Learning Engineer

Provider: Google Cloud

Issuer: Google Certifications

Completed / issued: 13 Feb 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://cloud.google.com/learn/certification/machine-learning-engineer>

IBM AI Engineering Professional Certificate

Provider: IBM

Issuer: IBM Training

Completed / issued: 21 May 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://www.ibm.com/training/badge/ibm-ai-engineering-professional-certificate>

IBM Certified watsonx Generative AI Engineer

Provider: IBM

Issuer: IBM Training

Completed / issued: 21 May 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://www.ibm.com/training/credentials/certifications>

NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL)

Provider: NVIDIA

Issuer: Nvidia Certification

Completed / issued: 21 June 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://www.nvidia.com/en-us/learn/certification/generative-ai-llm-associate/>

NVIDIA-Certified Professional: Generative AI LLMs (NCP-GENL)

Provider: NVIDIA

Issuer: Nvidia Certification

Completed / issued: 2 July 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://www.nvidia.com/en-us/learn/certification/generative-ai-llm-professional/>

AWS Certified Solutions Architect - Associate

Provider: AWS

Issuer: AWS Certifications

Completed / issued: 1 March 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://aws.amazon.com/certification/certified-solutions-architect-associate/>

AWS Certified Solutions Architect - Professional

Provider: AWS

Issuer: AWS Certifications

Completed / issued: 13 Feb 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://aws.amazon.com/certification/certified-solutions-architect-professional/>

Google Cloud Professional Cloud Architect

Provider: Google Cloud

Issuer: Google Certifications

Completed / issued: 1 Feb 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://cloud.google.com/learn/certification/cloud-architect>

Microsoft Certified: Cybersecurity Architect Expert (SC-100)

Provider: Microsoft

Issuer: Microsoft Learn

Completed / issued: 6 June 2025

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://learn.microsoft.com/en-us/credentials/certifications/cybersecurity-architect-expert/>

AWS Certified Security - Specialty

Provider: AWS

Issuer: AWS Certifications

Completed / issued: 3 Jan 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://aws.amazon.com/certification/certified-security-specialty/>

Google Cloud Professional Cloud Security Engineer

Provider: Google Cloud

Issuer: Google Certifications

Completed / issued: 13 July 2026

Validity: No expiration stated

Type: Course completion

Source: Official program page - <https://cloud.google.com/learn/certification/cloud-security-engineer>

Education, Technical Range and Contact

Education

Degree: Sarjana Teknik (S.T.), Informatics Engineering

Institution: Universitas Pancasila

Judicium: 22 February 2024

Diploma date: 16 May 2024

Technical Range

- Languages and APIs: Python, JavaScript, TypeScript, Node.js, FastAPI, REST
- Data: PostgreSQL, ClickHouse, MySQL, Oracle, SQL, ChromaDB, PGVector
- AI and LLM: LangChain, LangGraph, PyTorch, TensorFlow, Hugging Face, RAG, RLHF, fine-tuning, prompt engineering
- Platform: dbt-core, Airflow, Airbyte, OpenMetadata, Docker, vLLM, SGLang
- Analytics: Power BI, Power Query, VBA, statistical modeling and evaluation
- Security and quality: RBAC, audit logging, Playwright, red-teaming, guardrails, UAT, code review and information-security risk

Best-Fit Roles

- AI Architect / Generative AI Architect
- AI Engineering Manager / Lead AI Engineer
- LLM Engineer / LLM Systems Architect
- Applied AI Engineer
- AI Platform Engineer
- Data and AI Solutions Architect
- AI Security and LLM Red-Team Practitioner

Contact

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GitHub: <https://github.com/Faris-Maulana>

Portfolio: <https://faris-portfolio-red.vercel.app/>

LinkedIn: <https://linkedin.com/in/faris-maulana-0035b914a>